

## Bilinear Transform Derivation (Trapezoidal Integration)

Handwritten Derivation - DSP Exam Notes

Derivation of s-to-z Mapping via Numerical Integration:

Consider continuous 1st order differential equation:

$$dy(t)/dt = x(t)$$

Integrate both sides from  $nT$  to  $(n+1)T$ :

$$\int_{nT}^{(n+1)T} dy(t) = \int_{nT}^{(n+1)T} x(t) dt$$

$$y(nT + T) - y(nT) = \int_{nT}^{(n+1)T} x(t) dt$$

Apply Trapezoidal Rule of Numerical Integration:

$$y[n+1] - y[n] \approx (T/2) * [x[n+1] + x[n]]$$

Take Z-transform of both sides:

$$z Y(z) - Y(z) = (T/2) [z X(z) + X(z)]$$

$$Y(z)(z - 1) = (T/2) X(z)(z + 1)$$

$$H(z) = Y(z)/X(z) = (T/2) * ((z + 1)/(z - 1))$$

Comparing with continuous integration  $H_a(s) = 1/s$ :

$$1/s = (T/2) * ((z + 1)/(z - 1))$$

$$\Rightarrow s = (2/T) * ((z - 1)/(z + 1)) = (2/T) * ((1 - z^{-1})/(1 + z^{-1}))$$

Q.E.D. Standard Bilinear Transformation Formula.

## Handwritten Warping Analysis

Derivation: Mapping the Frequency Axis

Substitute  $s = j\Omega$  (analog) and  $z = e^{j\omega}$  (digital):

$$j\Omega = \frac{2}{T} \cdot \frac{(e^{j\omega} - 1)}{(e^{j\omega} + 1)}$$

Factor out  $e^{j\omega/2}$ :

$$j\Omega = \frac{2}{T} \cdot \frac{e^{j\omega/2} (e^{j\omega/2} - e^{-j\omega/2})}{(e^{j\omega/2} (e^{j\omega/2} + e^{-j\omega/2}))}$$

$$j\Omega = \frac{2}{T} \cdot \frac{2j \sin(\omega/2)}{2 \cos(\omega/2)}$$

$$j\Omega = j \cdot \frac{2}{T} \cdot \tan(\omega/2)$$

$$\implies \Omega = \frac{2}{T} \cdot \tan(\omega/2)$$

$$\text{Inversely: } \omega = 2 \cdot \arctan(\Omega \cdot T/2)$$

Key Observations from Plot:

- For small  $\omega$ ,  $\tan(\omega/2) \approx \omega/2 \implies \Omega \approx \omega/T$  (Linear).
- As  $\omega \rightarrow \pi$ ,  $\tan(\omega/2) \rightarrow \infty \implies$  entire infinite analog axis  $[-\infty, \infty]$  maps into  $[-\pi, \pi]$ .
- Frequency Warping: High analog frequencies are heavily compressed into the digital unit circle.

## Handwritten Butterworth Order Derivation

Step-by-Step Order N Formula Derivation:

Butterworth magnitude squared function:

$$|H(j\Omega)|^2 = 1 / ( 1 + (\Omega / \Omega_c)^{2N} )$$

At passband edge  $\Omega_p$  with max ripple  $\alpha_p$  (dB):

$$10 * \log_{10}( 1 + (\Omega_p / \Omega_c)^{2N} ) = \alpha_p$$

$$1 + (\Omega_p / \Omega_c)^{2N} = 10^{(0.1 * \alpha_p)}$$

$$(\Omega_p / \Omega_c)^{2N} = 10^{(0.1 * \alpha_p)} - 1 \quad \text{---- (Eq. 1)}$$

At stopband edge  $\Omega_s$  with min attenuation  $\alpha_s$  (dB):

$$(\Omega_s / \Omega_c)^{2N} = 10^{(0.1 * \alpha_s)} - 1 \quad \text{---- (Eq. 2)}$$

Divide Eq. 2 by Eq. 1:

$$(\Omega_s / \Omega_p)^{2N} = ( 10^{(0.1 * \alpha_s)} - 1 ) / ( 10^{(0.1 * \alpha_p)} - 1 )$$

Take log10 of both sides:

$$2N * \log_{10}(\Omega_s / \Omega_p) = \log_{10}( (10^{(0.1 * \alpha_s)} - 1) / (10^{(0.1 * \alpha_p)} - 1) )$$

$$N = \text{ceil}( \log_{10}( (10^{(0.1 * \alpha_s)} - 1) / (10^{(0.1 * \alpha_p)} - 1) ) / (2 * \log_{10}(\Omega_s / \Omega_p)) ).$$

## Handwritten Pole Placement Diagram

Pole Distribution for Butterworth Prototype:

For  $N = 3$  Butterworth Filter:

All poles lie on circle of radius  $\Omega_c$  in left-half  $s$ -plane.

Pole locations:  $s_k = \Omega_c \cdot e^{j \cdot \theta_k}$ , where:

$\theta_k = \pi/2 + (2k - 1) \cdot \pi / (2 \cdot N)$  for  $k = 1, 2, 3$ .

-  $k = 1$ :  $\theta_1 = \pi/2 + \pi/6 = 2\pi/3 = 120$  degrees

$s_1 = \Omega_c \cdot (-0.5 + j \cdot 0.866)$

-  $k = 2$ :  $\theta_2 = \pi/2 + 3\pi/6 = \pi = 180$  degrees

$s_2 = -\Omega_c$

-  $k = 3$ :  $\theta_3 = \pi/2 + 5\pi/6 = 4\pi/3 = 240$  degrees

$s_3 = \Omega_c \cdot (-0.5 - j \cdot 0.866)$

Prototype Transfer Function:

$H_a(s) = \Omega_c^3 / ((s - s_1)(s - s_2)(s - s_3)) = \Omega_c^3 / ((s + \Omega_c)(s^2 + \Omega_c s + \Omega_c^2))$ .

## Worked Problem: 2nd Order Butterworth Design (Part 1)

Handwritten Worked Example - Exam Problem:

Problem Statement: Design a 2nd-order digital lowpass Butterworth filter using BLT.

Specifications:

- Sampling Frequency:  $F_s = 10 \text{ kHz}$  ( $T = 0.0001 \text{ s} = 10^{-4} \text{ s}$ )
- Cutoff frequency:  $f_c = 1 \text{ kHz}$

Step 1: Calculate Digital Cutoff Frequency  $\omega_c$ :

$$\omega_c = 2 * \pi * f_c / F_s = 2 * \pi * 1000 / 10000 = 0.2 * \pi \text{ rad/sample.}$$

Step 2: Frequency Prewarping to Find Analog Cutoff  $\Omega_c$ :

$$\Omega_c = (2 / T) * \tan(\omega_c / 2)$$

$$\Omega_c = 2 * 10000 * \tan(0.1 * \pi)$$

$$\tan(0.1 * \pi) = \tan(18 \text{ degrees}) = 0.324919696$$

$$\Omega_c = 20000 * 0.32492 = 6498.39 \text{ rad/s.}$$

Step 3: Normalized 2nd Order Butterworth Analog Transfer Function:

$$H_a(s) = \Omega_c^2 / (s^2 + \sqrt{2} * \Omega_c * s + \Omega_c^2).$$

## Worked Problem: 2nd Order Butterworth Design (Part 2)

Handwritten Worked Example (Continued):

Step 4: Substitute Bilinear Transformation  $s = 20000 * (z - 1)/(z + 1)$ :

Let  $K = 20000$  and  $\Omega_c = 6498.39$  rad/s.

Denominator  $D(z) = [ K*(z - 1)/(z + 1) ]^2 + \sqrt{2}*\Omega_c*[ K*(z - 1)/(z + 1) ] + \Omega_c^2$ .

Multiply numerator and denominator by  $(z + 1)^2$ :

$$\text{Numerator} = \Omega_c^2 * (z + 1)^2 = \Omega_c^2 * (1 + 2z^{-1} + z^{-2}) z^2.$$

Denominator after collecting coefficients:

$$A = K^2 + \sqrt{2}*K*\Omega_c + \Omega_c^2 = 4*10^8 + 1.838*10^8 + 4.223*10^7 = 6.2603 * 10^8.$$

$$B = -2*K^2 + 2*\Omega_c^2 = -8*10^8 + 0.8446*10^8 = -7.1554 * 10^8.$$

$$C = K^2 - \sqrt{2}*K*\Omega_c + \Omega_c^2 = 4*10^8 - 1.838*10^8 + 4.223*10^7 = 2.5843 * 10^8.$$

Dividing through by A:

$$b_0 = \Omega_c^2 / A = 4.2229*10^7 / 6.2603*10^8 = 0.06745$$

$$a_1 = B / A = -7.1554 / 6.2603 = -1.1430$$

$$a_2 = C / A = 2.5843 / 6.2603 = 0.4128$$

Final Digital Transfer Function:

$$H(z) = ( 0.06745 * (1 + 2z^{-1} + z^{-2}) ) / ( 1 - 1.1430z^{-1} + 0.4128z^{-2} ).$$

## Handwritten Notes on Limit Cycles & Overflow

Finite Wordlength Notes - Practical Observations:

### 1. Zero-Input Limit Cycles:

In recursive filter  $y[n] = -a y[n-1] + x[n]$ :

If input  $x[n]$  goes to zero, rounding arithmetic can sustain persistent oscillations.

Remedy: Magnitude Truncation (always truncate towards zero) instead of rounding.

Deadband zone:  $[-1 / (2*(1 - |a|)), 1 / (2*(1 - |a|))]$ .

### 2. Overflow Limit Cycles:

When two large positive fixed-point numbers add together, two's complement wraps around to large negative number.

Causes full-scale high-energy square wave oscillation (overflow limit cycle).

Remedy: Saturation Arithmetic Logic - clamp any sum exceeding +1 to +0.9999 and any sum below -1 to -1.0.

# Handwritten Comparison Table: FIR vs IIR Filters

Master Exam Cheat Sheet - FIR vs IIR Summary Table:

| Feature               | FIR Filter                  | IIR Filter                    |
|-----------------------|-----------------------------|-------------------------------|
| Phase Linearity       | Exactly Linear Phase        | Non-linear phase              |
| Stability             | Always Stable (All poles=0) | Can become unstable           |
| Filter Order for Spec | High (Large delay/latency)  | Low (Sharp cutoff)            |
| Memory Requirements   | Higher                      | Lower                         |
| Implementation Form   | Direct Form, Polyphase      | Cascade / Parallel Biquads    |
| Limit Cycles          | No limit cycles             | Susceptible to limit cycles   |
| Design Methods        | Windowing, Parks-McClellan  | Bilinear Transform, Imp. Inv. |